NOTEPAD APPLICATION – PSEUDOCODE

SECTION 1: File I/O Operations

```
MODULE FileOperations


  FUNCTION openFile()
    Create file chooser filtered for "*.txt" files
    Show open file dialog box
    IF user selects a file THEN
      Set currentFile to selected file
      TRY
        Read text content from currentFile
        Display content in main text area
        Update window title with current file name
      CATCH file read error
        Display error popup ("Error opening file")
      END TRY
    END IF
  END FUNCTION
  FUNCTION saveFileAs()


    Create file chooser filtered for "*.txt" files
    Show save file dialog box
    IF user selects a file destination THEN
      Set currentFile to selected file
```

```
        IF file name does not end with ".txt" THEN
            Append ".txt" to file name
        END IF
        Call saveFile()
        Update window title with current file name
    END IF
  END FUNCTION
  FUNCTION newFile()
    IF main text area is not empty THEN

        Prompt user to save unsaved changes
        IF user chooses YES THEN
            Call saveFile()
        ELSE IF user chooses CANCEL THEN
            Stop and return to editor
        END IF
    END IF
    Clear main text area
    Set currentFile to empty/null
    Set window title to "Untitled"
  END FUNCTION
END MODULE
```

SECTION 2: Edit Actions & Shortcuts

```
MODULE EditActions

  FUNCTION initializeEditMenu()
    // Bind shortcut key accelerators to edit components
    Set itemCut accelerator to Ctrl + X
    Set itemCopy accelerator to Ctrl + C
    Set itemPaste accelerator to Ctrl + V
    Set itemSelectAll accelerator to Ctrl + A

    // Create "Clear All" menu item
    Create itemClearAll as new JMenuItem("Clear All")

    Add action listener to itemClearAll:
      Show confirmation dialog: "Are you sure you want to clear all text?"
      IF user clicks YES option THEN
        Set main text area text to empty ("")
      END IF
    END action listener

    // Add components to the Edit Menu
    Add separator line to editMenu
    Add itemClearAll to editMenu
  END FUNCTION
```

END MODULE

SECTION 3: Font Selection & Formatting

```
MODULE FontFormatting


  FUNCTION showFontDialog(currentFont)
    Get list of available system fonts
    Prompt user to choose a font family from the list
    IF user selects a font family THEN


      Return new Font(selected family, current style, current size)
    ELSE
      Return currentFont
    END IF
  END FUNCTION
  FUNCTION changeFontSize()
    Call showFontDialog with current text font
    Apply returned font to main text area
  END FUNCTION
END MODULE
```

SECTION 4: Status Bar & View Menu

```
MODULE StatusBar

  FUNCTION updateStatusBar()



    Get current cursor/caret position
    Calculate line number and column number
    Get text from main text area
    Calculate total character count

    Calculate character count without spaces
    Calculate word count by splitting text on spaces
    Update status bar label:
      "Line: X | Col: Y | Words: W | Chars: C"
  END FUNCTION
  FUNCTION toggleStatusBar(isVisible)
    Set status bar visibility to isVisible
  END FUNCTION
END MODULE
```

SECTION 5: Unit Testing Rules

```
MODULE UnitTesting
  TEST testWordCount()
    Given text = "The quick brown fox jumps over the lazy dog"

    Count words by splitting text on whitespace

    ASSERT count equals 9
  END TEST
  TEST testEmptyTextCount()
    Given text = ""

    ASSERT count equals 0
  END TEST
END MODULE
```